

1.4 - Inverse; Algebraic Props. of Matrices

Properties of Matrix Arithmetic:-

1. $A + B = B + A$ (commutative)
2. $A + (B + C) = (A + B) + C$ (ASSOCIATIVE)
3. $A(B + C) = AB + AC$ Left Dist. Law
4. $(A + B)C = AC + BC$
5. $A(BC) = (AB)C$ ASSOCIATIVITY for matrix multiplication

Example: 2- (order matters in Matrix Multiplication)

$$A = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$$

$AB = ?$

$$AB = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} -1+0 & -2+0 \\ 2+9 & 4+0 \end{bmatrix}$$

$$AB = \begin{bmatrix} -1 & -2 \\ 11 & 4 \end{bmatrix}$$

$BA = ?$

$$BA = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix}$$

$$BA = \begin{bmatrix} -1+4 & 0+6 \\ -3+0 & 0+0 \end{bmatrix}$$

$$BA = \begin{bmatrix} 3 & 6 \\ -3 & 0 \end{bmatrix}$$

So $AB \neq BA$

Example: 3 (Failure of Cancellation Law)

For Real numbers:-

$$a \cdot b = a \cdot c \quad \text{and} \quad a \neq 0 \Rightarrow b = c$$

But For Matrices:-

$$A \cdot B = A \cdot C \Rightarrow B = C$$

↳ Not always True.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$$

$A \cdot B = ?$

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0+3 & 0+4 \\ 0+6 & 0+8 \end{bmatrix}$$

$A \cdot C = ?$

$$AC = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$$

$$AC = \begin{bmatrix} 0+3 & 0+4 \\ 0+6 & 0+8 \end{bmatrix}$$

$$AB = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

$$AC = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

So, $AB = AC$ but $B \neq C$

Example: 4 (zero Product with non zero Factors)

↳ For numbers, $ab = 0 \Rightarrow a = 0$ or $b = 0$

↳ For matrices, $AB = 0$ doesn't imply $A = 0$ or $B = 0$

↳ Imp:- Matrices product can be zero if both matrices are not even zero unlike real numbers.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 3 & 7 \\ 0 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 7 \\ 0 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0+0 & 0+0 \\ 0+0 & 0+0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

→ Here $AB = 0$ even $A \neq 0$
and $B \neq 0$

Example: 5 (An Invertible Matrix)

→ A square matrix A is invertible if there exist another square matrix B of same size such that

$$AB = BA = I$$

Where I is Identity Matrix.

• B is called inverse of A written as $B = A^{-1}$

• If no such matrix exist then A is a singular. (Non Inv)

Remark:

If $AB = BA = I$, then

- A is inverse of B
- B is inverse of A

$$A = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}, B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$$

$AB = ?$

$$AB = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 6-5 & 10-10 \\ -3+3 & -5+6 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$BA = ?$

$$BA = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}$$

$$BA = \begin{bmatrix} 6-5 & -15+15 \\ 2-2 & -5+6 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Since

$$\rightarrow AB = I \text{ and } BA = I$$

$\rightarrow$ A is invertible and B is invertible

$\rightarrow$ A and B are inverse of each other.

Example: 6 (A Class of Singular Matrix)

A square with a row or column of zero is singular. It doesn't have inverse (non invertible) coz we multiply singular matrix with any other matrix the resultant will not be identity.

Idea: $\rightarrow$ To show A a singular. We must prove that there is no such B that $AB = BA = I$

$$A = \begin{bmatrix} 1 & 4 & 0 \\ 2 & 5 & 0 \\ 3 & 6 & 0 \end{bmatrix}$$

↳ Singular (Non-Invertible)

Symbol:

$$\begin{aligned} &\downarrow \\ A(A^{-1}) &= I \\ AA^{-1} &= I = A^{-1}A \end{aligned}$$

- AA^{-1} = Matrix A is multiplied by its inverse A^{-1} , result is identity matrix
- $A^{-1}A$ = When inverse of A is multiplied by A, then result is also identity.

Example: 7 (Calculating inverse of 2x2 Matrix)

a-

$$A = \begin{bmatrix} 6 & 1 \\ 5 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det A} \text{Adj } A$$

$$\det A = ?$$

$$\det A = (6)(2) - (5)(1)$$

$$\det A = 12 - 5$$

$$\det A = 7$$

$$\text{Adj } A = ?$$

$$\text{Adj } A = \begin{bmatrix} 2 & -1 \\ -5 & 6 \end{bmatrix}$$

b-

$$A = \begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det A} \text{Adj } A$$

$$\det A = ?$$

$$\det A = (-1)(-6) - (3)(2)$$

$$\det A = 6 - 6$$

$$\det A = 0$$

↳ A is not invertible because $\det A = 0$

$$A^{-1} = ?$$

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & -1 \\ -5 & 6 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 2/7 & -1/7 \\ -5/7 & 6/7 \end{bmatrix} \rightarrow \text{Ans.}$$

Inverse Properties:

$$1. A^0 = I$$

$$2. A \cdot A = A^2$$

$$3. A^k = A \cdot A \cdot A \dots A \rightarrow k\text{-Times.}$$

$$4. A^{-3} = (A^{-1})^3 = (A^{-1})(A^{-1})(A^{-1})$$

$$5. A^{-k} = (A^{-1})^k = (A^k)^{-1}$$

$$6. (AB)^{-1} = B^{-1}A^{-1} \rightarrow \text{Sock Shoe Prop}$$

$$7. (ABC)^{-1} = C^{-1}B^{-1}A^{-1}$$

$$8. (A^{-1})^{-1} = A$$

$$9. (kA)^{-1} = \frac{1}{k} A^{-1}$$

Some Formula

In Algebra:

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$= (a+b)(a+b)$$

$$= a^2 + ab + ba + b^2$$

$$= a^2 + 2ab + b^2$$

In Matrix:-

$$1. (A+B)^2 \neq A^2 + 2AB + B^2$$

$$\rightarrow AB \neq BA$$

$$2. (A-B)^2 \neq A^2 - 2AB + B^2$$

Example : 9

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}, B = \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\text{verify } (AB)^{-1} = B^{-1}A^{-1}$$

$$(AB)^{-1} = ? \quad AB^{-1} = \frac{1}{\det AB} \text{adj } AB$$

$$AB = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix} \Rightarrow AB = \begin{bmatrix} 3+4 & 2+4 \\ 3+6 & 2+6 \end{bmatrix}$$

$$AB = \begin{bmatrix} 7 & 6 \\ 9 & 8 \end{bmatrix}$$

det AB

$$\det AB = (-7)(8) - (9)(6)$$

$$\det AB = 56 - 54$$

$$\det AB = 2$$

adj A

$$\text{adj } A = \begin{bmatrix} 8 & -6 \\ -9 & 7 \end{bmatrix}$$

$$(AB)^{-1} = ?$$

$$(AB)^{-1} = \frac{1}{2} \begin{bmatrix} 8 & -6 \\ -9 & 7 \end{bmatrix}$$

$$(AB)^{-1} = \begin{bmatrix} 4 & -3 \\ -9/2 & 7/2 \end{bmatrix}$$

$$B^{-1} A^{-1} = ?$$

$$B^{-1} = \frac{1}{\det B} \text{adj } B$$

$$\det B = ?$$

$$\det B = (3)(2) - (2)(2)$$

$$\det B = 6 - 4$$

$$\det B = 2$$

$$B^{-1} = ?$$

$$B^{-1} = \frac{1}{2} \begin{bmatrix} 2 & -2 \\ -2 & 3 \end{bmatrix}$$

$$B^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 3/2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

$$\det A = ?$$

$$\det A = (1)(3) - (1)(2)$$

$$\det A = 3 - 2$$

$$\det A = 1$$

$$A^{-1} = ?$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

$$B^{-1} A^{-1} = ?$$

$$B^{-1} A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 3/2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

$$B^{-1}A^{-1} = \begin{bmatrix} 3+1 & -2-1 \\ -3-\frac{3}{2} & 2+\frac{3}{2} \end{bmatrix}$$

$$\frac{-3-\frac{3}{2}}{2} = \frac{-6-3}{2} = \frac{-9}{2}$$

$$\frac{2+\frac{3}{2}}{2} = \frac{4+1}{2} = \frac{5}{2}$$

$$B^{-1}A^{-1} = \begin{bmatrix} 4 & -3 \\ -\frac{9}{2} & \frac{5}{2} \end{bmatrix}$$

$$\text{so } (AB)^{-1} = B^{-1}A^{-1}$$

Exercise Set: 1.4

Question: 1-2

verify the following matrix and scalars satisfy stated properties of Theorem 1.4.1.

$$A = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$a = 4, \quad b = -7$$

Question: 1

or

$$A + (B + C) = (A + B) + C \quad (\text{ASSO Law For matrix addition})$$

$$A + (B + C) = ?$$

$$(A + B) + C = ?$$

$$A + (B + C) = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} + \left(\begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} \right) \quad (A + B) + C = \left(\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix} \right) + \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$A + (B + C) = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 3 \\ -2 & -6 \end{bmatrix}$$

$$(A + B) + C = \begin{bmatrix} 3 & 1 \\ 3 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$A + (B + C) = \begin{bmatrix} 7 & 2 \\ 0 & -2 \end{bmatrix}$$

$$(A + B) + C = \begin{bmatrix} 7 & 2 \\ 0 & -2 \end{bmatrix}$$

So $A + (B + C) = (A + B) + C$

b-

ASSO. Law For multiplication $A(BC) = (AB)C$

$A(BC) = ?$

$$= \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \left(\begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \left(\begin{bmatrix} 0-6 & 0-4 \\ 4+12 & 1+8 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -6 & -4 \\ 16 & 9 \end{bmatrix} \Rightarrow \begin{bmatrix} -18-16 & -12-9 \\ -12+64 & -8+36 \end{bmatrix}$$

$$= \begin{bmatrix} -34 & -21 \\ 52 & 28 \end{bmatrix}$$

$(AB)C = ?$

$$= \left(\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix} \right) \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$= \left(\begin{bmatrix} 0-1 & 6+4 \\ 0+4 & 4-16 \end{bmatrix} \right) \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 10 \\ 4 & -12 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -4-30 & -1-20 \\ 16+36 & 4-24 \end{bmatrix}$$

$$= \begin{bmatrix} -34 & -21 \\ 52 & 28 \end{bmatrix}$$

So $A(BC) = (AB)C$

c-

Left Dist. Law

$$A(B+C) = AB+AC$$

d-

$$(a+b)C = aC + bC$$

$$(a+b)C = ?$$

$$= (4-7) \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} \Rightarrow (-3) \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -12 & -3 \\ 9 & 6 \end{bmatrix}$$

$$aC + bC = ?$$

$$aC = ?$$

$$= 4 \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 16 & 4 \\ -12 & -8 \end{bmatrix}$$

$$bC = ?$$

$$= -7 \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -28 & -7 \\ 21 & 14 \end{bmatrix}$$

$$aC + bC = ?$$

$$= \begin{bmatrix} 16 & 4 \\ -12 & -8 \end{bmatrix} + \begin{bmatrix} -28 & -7 \\ 21 & 14 \end{bmatrix}$$

$$= \begin{bmatrix} -12 & -3 \\ 9 & 6 \end{bmatrix}$$

$$\text{so, } (a+b)C = aC + bC$$

Question: 02

$$(a) a(BC) = (aB)C = B(aC)$$

$$(b) A(B-C) = AB-AC =$$

$$(c) (B+C)A = BA+CA$$

$$(d) a(bc) = (ab)c$$

use matrix and values from previous question and apply simple operation to both sides and check whether they come as equal or not. If equal then verified and if not equal then not verified.

Question: 3-4

matrix and values from previous question.

3-(a)

$$(A^T)^T = \left(\begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}^T \right)^T$$

$$(A^T)^T = \left(\begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix} \right)^T$$

$$(A^T)^T = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$$

$$\text{so } (A^T)^T = A$$

3-(b)

$$(AB)^T = B^T A^T$$

$$(AB)^T = ?$$

$$AB = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0-1 & 6+4 \\ 0+4 & 4-16 \end{bmatrix}$$

$$B^T = ?$$

$$B^T = \begin{bmatrix} 0 & 1 \\ 2 & -4 \end{bmatrix}$$

$$A^T = ?$$

$$A^T = \begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix}$$

$$(AB)^T = \begin{bmatrix} -1 & 10 \\ 4 & -12 \end{bmatrix}^t$$

$$B^T A^T = \begin{bmatrix} 0 & 1 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix}$$

$$(AB)^T = \begin{bmatrix} -1 & 4 \\ 10 & -12 \end{bmatrix}$$

$$B^T A^T = \begin{bmatrix} 0-1 & 0+4 \\ 6+4 & 4-16 \end{bmatrix}$$

$$B^T A^T = \begin{bmatrix} -1 & 4 \\ 10 & -12 \end{bmatrix}$$

so $(AB)^T = B^T A^T$

4 (a) -

$$(A+B)^T = A^T + B^T$$

$$(A+B)^T = ?$$

$$A+B = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 3 & 1 \\ 3 & 0 \end{bmatrix}$$

$$(A+B)^T = \begin{bmatrix} 3 & 1 \\ 3 & 0 \end{bmatrix}^t$$

$$(A+B)^T = \begin{bmatrix} 3 & 3 \\ 1 & 0 \end{bmatrix}$$

so $(A+B)^T = A^T + B^T$

$$A^T = ?$$

$$A^T = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}^t = \begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix}$$

$$B^T = ?$$

$$B^T = \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix}^t = \begin{bmatrix} 0 & 1 \\ 2 & -4 \end{bmatrix}$$

$$A^T + B^T$$

$$A^T + B^T = \begin{bmatrix} 3 & 2 \\ -1 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 2 & -4 \end{bmatrix}$$

$$A^T + B^T = \begin{bmatrix} 3 & 3 \\ 1 & 0 \end{bmatrix}$$

4 (b) -

$$(ac)^T = ac^T$$

$$(ac)^T = ?$$

$$(ac)^T = \left(4 \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix} \right)^T$$

$$ac^T = ?$$

$$ac^T = 4 \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}^T$$

$$(ac)^T = \begin{bmatrix} 16 & 4 \\ -12 & -8 \end{bmatrix}^T \quad \left| \quad ac^T = 4 \begin{bmatrix} 4 & -3 \\ 1 & -2 \end{bmatrix} \right.$$

$$(ac)^T = \begin{bmatrix} 16 & -12 \\ 4 & -8 \end{bmatrix} \quad \left| \quad ac^T = \begin{bmatrix} 16 & -12 \\ 4 & -8 \end{bmatrix} \right.$$

so $(ac)^T = ac^T$

Question: 5-8

Use Theorem 1.4.5 to compute inverse of the matrix.

Question: 5

$$A = \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A$$

$$\det A: ?$$

$$\operatorname{adj} A = ?$$

$$\det A = (2)(4) - (4)(-3)$$

$$\det A = 8 + 12$$

$$\det A = 20$$

→

$$\operatorname{adj} A = \begin{bmatrix} 4 & 3 \\ -4 & 2 \end{bmatrix}$$

$$A^{-1} = ?$$

$$A^{-1} = \frac{1}{20} \begin{bmatrix} 4 & 3 \\ -4 & 2 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{3}{20} \\ -\frac{1}{5} & \frac{1}{10} \end{bmatrix} \text{ - Ans}$$

Question: 6

$$B = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$$

same as Question: 5

Question: 8

b-

$$D = \begin{bmatrix} 6 & 4 \\ -2 & -1 \end{bmatrix}$$

$$D^{-1} = \frac{1}{\det D} \text{adj } D$$

$$\det D = ?$$

$$\det D = (6)(-1) - (-2)(4) \rightarrow$$

$$\det D = -6 + 8$$

$$\det D = 2$$

$$\text{adj } D = ?$$

$$\text{adj } D = \begin{bmatrix} -1 & -4 \\ 2 & 6 \end{bmatrix}$$

$$D^{-1} = ?$$

$$D^{-1} = \frac{1}{2} \begin{bmatrix} -1 & -4 \\ 2 & 6 \end{bmatrix} \Rightarrow D^{-1} = \begin{bmatrix} -\frac{1}{2} & -2 \\ 1 & 3 \end{bmatrix}$$

Question: 7

$$C = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

same as Question: 8

Question: 9:-

Find inverse of

$$\begin{bmatrix} \frac{1}{2}(e^x + e^{-x}) & \frac{1}{2}(e^x - e^{-x}) \\ \frac{1}{2}(e^x - e^{-x}) & \frac{1}{2}(e^x + e^{-x}) \end{bmatrix}$$

$$\text{Let } A^{-1} = \frac{1}{\det A} \text{adj } A$$

$$\det A =$$

$$\det A = \frac{1}{2}(e^x + e^{-x}) \frac{1}{2}(e^x + e^{-x}) - \frac{1}{2}(e^x - e^{-x}) \frac{1}{2}(e^x - e^{-x})$$

$$\det A = \frac{1}{4}(e^x + e^{-x})^2 - \frac{1}{4}(e^x - e^{-x})^2$$

$$\det A = \frac{1}{4}(e^{2x} + 2(e^x)(e^{-x}) + e^{-2x}) - \frac{1}{4}[e^{2x} - 2(e^x)(e^{-x}) + e^{-2x}]$$

$$\det A = \frac{1}{4}(e^{2x} + 2 + e^{-2x}) - \frac{1}{4}[e^{2x} - 2 + e^{-2x}]$$

$$\det A = \frac{e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}}{4} = \frac{4}{4}$$

$$\det A = 1$$

$$\text{adj } A = ?$$

$$\text{adj } A = \begin{bmatrix} \frac{1}{2}(e^x + e^{-x}) & -\frac{1}{2}(e^x - e^{-x}) \\ -\frac{1}{2}(e^x - e^{-x}) & \frac{1}{2}(e^x + e^{-x}) \end{bmatrix}$$

$$\text{adj } A = \begin{bmatrix} \frac{1}{2}(e^x + e^{-x}) & \frac{1}{2}(e^{-x} - e^x) \\ \frac{1}{2}(e^{-x} - e^x) & \frac{1}{2}(e^x + e^{-x}) \end{bmatrix}$$

$$A^{-1} = ?$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} \frac{1}{2}(e^x + e^{-x}) & \frac{1}{2}(e^{-x} - e^x) \\ \frac{1}{2}(e^{-x} - e^x) & \frac{1}{2}(e^x + e^{-x}) \end{bmatrix} \quad \text{Ans}$$

Question: 10.

Find inverse of

$$\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$\text{Let } A = \frac{1}{\det A} \text{adj } A$$

$$\det A = ?$$

$$\det A = (\cos \theta)(\cos \theta) - (-\sin \theta)(\sin \theta)$$

$$\det A = \cos^2 \theta + \sin^2 \theta$$

$$\det A = 1$$

$$A^{-1} = ?$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad \text{Ans}$$

$$\text{adj } A = ?$$

$$\text{adj } A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Question: 11-14

Verify the equations are valid for the matrices in exercises 5-8.

$$11. (A^T)^{-1} = (A^{-1})^T$$

$$12. (A^{-1})^{-1} = A$$

$$13. (ABC)^{-1} = C^{-1} B^{-1} A^{-1}$$

$$14. (ABC)^T = C^T B^T A^T$$

Easy. Use matrices from question 5-8 and apply simple operations to verify both sides.

Question: 15-18

Use given info to Find A

15-

$$(7A)^{-1} = \begin{bmatrix} -3 & 7 \\ 1 & -2 \end{bmatrix}$$

By Taking inverse of b/s

$$((7A)^{-1})^{-1} = \begin{bmatrix} -3 & 7 \\ 1 & -2 \end{bmatrix}^{-1}$$

$$7A = \frac{1}{-1} \begin{bmatrix} -2 & -7 \\ -1 & -3 \end{bmatrix}$$

$$7A = \begin{bmatrix} 2 & 7 \\ 1 & 3 \end{bmatrix}$$

$$A = \begin{bmatrix} \frac{2}{7} & 1 \\ \frac{1}{7} & \frac{3}{7} \end{bmatrix} \text{ Ans.}$$

$$\det A = (-3)(-2) - (1)(7)$$

$$\det A = 6 - 7$$

$$\det A = -1$$

$$\text{adj } A = \begin{bmatrix} -2 & -7 \\ -1 & -3 \end{bmatrix}$$

$$(A^{-1})^{-1} = A$$

Extra::

$$(-7A^{-1}) = \begin{bmatrix} -3 & 7 \\ 1 & -2 \end{bmatrix}$$

$$\therefore (KA)^{-1} = \frac{1}{k} A^{-1}$$

By taking inverse on b/s

$$(-7A^{-1})^{-1} = \begin{bmatrix} -3 & 7 \\ 1 & -2 \end{bmatrix}^{-1}$$

$$\frac{1}{-7} (A^{-1})^{-1} = \begin{bmatrix} 2 & +7 \\ +1 & 3 \end{bmatrix} \quad \text{From previous Qs. } (KA)$$

$$\frac{1}{-7} A = \begin{bmatrix} 2 & 7 \\ 1 & 3 \end{bmatrix}$$

$$(A^{-1})^{-1} = A$$

$$A = \begin{bmatrix} 14 & 49 \\ 7 & 21 \end{bmatrix} \quad \text{Ans}$$

Question: 16

$$(5A^T)^{-1} = \begin{bmatrix} -3 & -1 \\ 5 & 2 \end{bmatrix}$$

$$\det = (-3)(2) - (5)(-1)$$

$$\det = -6 + 5 = -1$$

Taking inverse of b/s

$$((5A^T)^{-1})^{-1} = \begin{bmatrix} -3 & -1 \\ 5 & 2 \end{bmatrix}^{-1}$$

$$\text{adj } A = \begin{bmatrix} 2 & 1 \\ -5 & -3 \end{bmatrix}$$

$$(5A^T) = \begin{bmatrix} -2 & -1 \\ 5 & 3 \end{bmatrix}$$

Taking Transpose on b/s

$$(5A^T)^T = \begin{bmatrix} -2 & -1 \\ 5 & 3 \end{bmatrix}^T$$

$$5(A^T)^T = \begin{bmatrix} -2 & 5 \\ -1 & 3 \end{bmatrix} \Rightarrow 5A = \begin{bmatrix} -2 & 5 \\ -1 & 3 \end{bmatrix}$$

$$A = \begin{bmatrix} -\frac{2}{5} & 1 \\ -\frac{1}{5} & \frac{3}{5} \end{bmatrix} \text{ Ans}$$

$$(KA^T)^T = KA$$

Question: 17

$$(I + 2A)^{-1} = \begin{bmatrix} -1 & 2 \\ 4 & 5 \end{bmatrix}$$

Taking inverse on b/s

$$((I + 2A)^{-1})^{-1} = \begin{bmatrix} -1 & 2 \\ 4 & 5 \end{bmatrix}^{-1}$$

$$\text{adj} = \begin{bmatrix} 5 & -2 \\ -4 & -1 \end{bmatrix}$$

$$\det = (-1)(5) - (4)(2)$$

$$\det = -5 - 8$$

$$\det = -13$$

$$I + 2A = \begin{bmatrix} -\frac{5}{13} & \frac{2}{13} \\ \frac{4}{13} & \frac{1}{13} \end{bmatrix}$$

$$-\frac{5}{13} - 1 = \frac{-5-13}{13} = \frac{-18}{13}$$

$$\frac{2}{13} - 0 = \frac{2}{13}$$

$$\frac{4}{13} - 0 = \frac{4}{13}$$

$$\frac{1}{13} - 1 = \frac{1-13}{13} = \frac{-12}{13}$$

$$2A = \begin{bmatrix} -\frac{5}{13} & \frac{2}{13} \\ \frac{4}{13} & \frac{1}{13} \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$2A = \begin{bmatrix} -\frac{18}{13} & \frac{2}{13} \\ \frac{4}{13} & -\frac{12}{13} \end{bmatrix}$$

$$A = \begin{bmatrix} -\frac{9}{13} & \frac{1}{13} \\ \frac{2}{13} & -\frac{6}{13} \end{bmatrix} \text{ Ans}$$

$$\det = 10 + 3 = 13$$

Question: 18

$$A^{-1} = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}$$

$$\text{adj} = \begin{bmatrix} 5 & 1 \\ -3 & 2 \end{bmatrix}$$

Taking inverse on b/s

$$(A^{-1})^{-1} = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}^{-1} \Rightarrow A = \begin{bmatrix} \frac{5}{13} & \frac{1}{13} \\ -\frac{3}{13} & \frac{2}{13} \end{bmatrix} \text{ Ans}$$

Question: 19-20

Compute following using given matrix A.

19-

$$A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

(a) A^3

$$A^3 = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 9+2 & 3+1 \\ 6+2 & 2+1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 33+8 & 11+4 \\ 24+6 & 8+3 \end{bmatrix} \Rightarrow \begin{bmatrix} 41 & 15 \\ 30 & 11 \end{bmatrix} \quad \text{Ans}$$

(b) $A^{-3} = (A^{-1})^3$

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$(A^{-1})^3 = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$(A^{-1})^3 = \begin{bmatrix} 1+2 & -1-3 \\ -2-6 & 2+9 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$(A^{-1})^3 = \begin{bmatrix} 3 & -4 \\ -8 & 11 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$(A^{-1})^3 = \begin{bmatrix} 3+8 & -3-12 \\ -8-22 & 8+33 \end{bmatrix}$$

$$\det A = (3)(1) - (2)(1)$$

$$\det A = 3 - 2$$

$$\det A = 1$$

$$\text{adj } A = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$(A^{-1})^3 = \begin{bmatrix} 11 & -15 \\ -30 & 41 \end{bmatrix} \text{ Ans}$$

$$(C) A^2 - 2A + I$$

$$A^2 = ?$$

$$A^2 = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 9+2 & 3+1 \\ 6+2 & 2+1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix}$$

$$2A = ?$$

$$2A = 2 \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$2A = \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix}$$

$$A^2 - 2A + I = ? \quad (A^2 + I) - 2A$$

$$= \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix} - \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & 4 \\ 8 & 4 \end{bmatrix} - \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix} \text{ Ans}$$

20-

$$20. A = \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix}$$

as q.19
same procedure will be used just
matrix is different

Question: 21-22

compute $p(A)$ for given matrix A and following polynomials.

21-

$$A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

(a) $p(x) = x - 2$

So $p(A) = A - 2I$

$$p(A) = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow p(A) = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}$$

(b) $p(x) = 2x^2 - x + 1$

So $p(A) = 2A^2 - A + I$

$$p(A) = 2 \left(\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \right) - \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$p(A) = 2 \left(\begin{bmatrix} 9+2 & 3+1 \\ 6+2 & 2+1 \end{bmatrix} \right) - \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 22 & 8 \\ 16 & 6 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 22 & 8 \\ 16 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 23 & 8 \\ 16 & 7 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 20 & 7 \\ 14 & 6 \end{bmatrix} \text{ Ans.}$$

$$(6) p(x) = x^3 - 2x + 1$$

$$\text{So } p(A) = A^3 - 2A + I$$

$$p(A) = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} - 2 \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 11 & 4 \\ 8 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 33+8 & 11+4 \\ 24+6 & 8+3 \end{bmatrix} - \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 41 & 15 \\ 30 & 11 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 42 & 15 \\ 30 & 12 \end{bmatrix} - \begin{bmatrix} 6 & 2 \\ 4 & 2 \end{bmatrix}$$

$$p(A) = \begin{bmatrix} 36 & 13 \\ 26 & 10 \end{bmatrix} \text{ Ans}$$

22-

$$A = \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix}$$

→ same method and eqn as of
q:21 just matrix different

Question: 23-24.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

23-

Find all values of a, b, c and d (if any) for which matrices A and B are commutative.

$$AB = BA \rightarrow (1)$$

$$AB = ?$$

$$BA = ?$$

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$BA = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$AB = \begin{bmatrix} 0+a & a+0 \\ 0+c & c+0 \end{bmatrix}$$

$$BA = \begin{bmatrix} 0+c & 0+d \\ 0+0 & 0+0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix}$$

$$BA = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix}$$

put in (1)

$$\begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix} = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} c=0 \\ a=d \end{bmatrix}$$

$$, \begin{bmatrix} b = \text{any value} \end{bmatrix} \text{ (since it disappears).}$$

24

Find all values of a, b, c and (if any) for which matrices A and C are commutative.

$$AC = CA \rightarrow (1)$$

$$AC = ?$$

$$CA = ?$$

$$AC = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$CA = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$AC = \begin{bmatrix} 0+b & 0+0 \\ 0+d & 0+0 \end{bmatrix}$$

$$AC = \begin{bmatrix} b & 0 \\ d & 0 \end{bmatrix}$$

$$CA = \begin{bmatrix} 0+0 & 0+0 \\ a+0 & b+0 \end{bmatrix}$$

$$CA = \begin{bmatrix} 0 & 0 \\ a & b \end{bmatrix}$$

put in (1)

$$\begin{bmatrix} b & 0 \\ d & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ a & b \end{bmatrix}$$

$$\boxed{b=0}, \boxed{a=d}$$

$$\boxed{c = \text{can be any value}}$$

Question: 025-28-

$$AX=B$$

Use matrix inversion
solution of given system $X=A^{-1}B$ to find unique

26-

$$\begin{aligned} -x_1 + 5x_2 &= 4 \\ -x_1 - 3x_2 &= 1 \end{aligned}$$

$$AX=B$$

$$\begin{bmatrix} -1 & 5 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ -1 & -3 \end{bmatrix}^{-1} \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$A^{-1}=?$$

$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

$$\det A = (-1)(-3) - (-1)(5)$$

$$\det A = 3 + 5 = 8$$

$$\text{adj } A = \begin{bmatrix} -3 & 5 \\ 1 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{8} \begin{bmatrix} -3 & 5 \\ 1 & -1 \end{bmatrix}$$

$$X = A^{-1}B$$

$$X = \frac{1}{8} \begin{bmatrix} -3 & 5 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} \Rightarrow \frac{1}{8} \begin{bmatrix} -12 + 5 \\ 4 - 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -17/8 \\ 3/8 \end{bmatrix}$$

$$\boxed{x_1 = -17/8}, \boxed{x_2 = 3/8}$$

25

$$3x_1 - 2x_2 = -1$$

$$4x_1 + 5x_2 = 3$$

27

$$6x_1 + x_2 = 0$$

$$4x_1 - 3x_2 = -2$$

28

$$2x_1 - 2x_2 = 4$$

$$x_1 + 4x_2 = 4$$

↳ Same as Q: 26.

True-False Exercises

True-False Exercises

(a) Two $n \times n$ matrices, A and B , are inverses of one another if and only if $AB = BA = 0$.

- **Answer: False**
- **Reason:** A and B are inverses if $AB = BA = I$, not 0. If $AB = 0$, then at least one of the matrices must be singular.

(b) For all square matrices A and B of the same size, $(A + B)^2 = A^2 + 2AB + B^2$.

- **Answer: False**
- **Reason:** Matrix multiplication is **not commutative** in general.

$$(A + B)^2 = A^2 + AB + BA + B^2 \neq A^2 + 2AB + B^2$$

Only true if $AB = BA$.

(c) For all square matrices A and B of the same size, $A^2 - B^2 = (A - B)(A + B)$.

- **Answer: False**
- **Reason:** For matrices:

$$(A - B)(A + B) = A^2 + AB - BA - B^2 \neq A^2 - B^2$$

Only true if $AB = BA$.

(d) If A and B are invertible matrices of the same size, then AB is invertible and $(AB)^{-1} = A^{-1}B^{-1}$.

- **Answer: False**
- **Reason:** Correct formula:

$$(AB)^{-1} = B^{-1}A^{-1} \neq A^{-1}B^{-1}$$

(e) If AB is defined, then $(AB)^T = A^T B^T$.

- **Answer:** False
- **Reason:** Correct property:

$$(AB)^T = B^T A^T$$

(f) The matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible if and only if $ad - bc \neq 0$.

- **Answer:** True
- **Reason:** This is the standard formula for a 2×2 matrix determinant.

(g) If A and B are matrices of the same size and k is a constant, then $(kA + B)^T = kA^T + B^T$.

- **Answer:** True
- **Reason:** Transpose distributes over addition and scalar multiplication.

(h) If A is an invertible matrix, then so is A^T .

- **Answer:** True
- **Reason:** $(A^T)^{-1} = (A^{-1})^T$

(j) A square matrix containing a row or column of zeros cannot be invertible.

- **Answer:** True
- **Reason:** Determinant = 0 $\rightarrow$ matrix is singular.

(k) The sum of two invertible matrices of the same size must be invertible.

- **Answer:** False
- **Reason:** Counterexample:

$$A = I, \quad B = -I \implies A + B = 0 \text{ (not invertible)}$$